Computer Vision and Signal Processing for ROI Extraction and Pulse Estimation

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1. Project Overview

This project focuses on extracting facial regions of interest from patient video data and estimating pulse rate from forehead and sclera video signals. The work consists of two main parts: a computer vision pipeline for region extraction and a signal processing pipeline for pulse estimation.

The computer vision part aims to detect and extract the forehead, left eye, and right eye regions from video data of five patients. Each extracted region is stored separately in binary format. The original task required accurate ROI extraction, consistent region dimensions, image quality control, and binary storage of each region.

The signal processing part focuses on estimating pulse rate from binary video data located in forehead and sclera folders. The provided video data has RGB channels, a frame rate of 60 fps, and variable frame lengths. The task required decoding binary video files, extracting temporal signals, applying filtering and frequency analysis, and reporting BPM values.

2. Project Structure

The project is organized under two main sections:

Task1: Computer Vision:

I. ROI extraction codes
II. Binary file generation codes
III. Example visual outputs

Task2: Signal Processing:

I. Binary video reading codes
II. Filtering and frequency analysis codes
III. BPM estimation outputs

3. Task 1 – Computer Vision: ROI Extraction

The first part of the project deals with extracting facial regions of interest from patient video data. The target regions are forehead region, left eye region, and right eye region

The main objective was to detect these regions accurately and store them in a reusable binary format. The extraction process was designed to avoid incomplete crops, misaligned regions, or incorrect facial areas. Image quality was also considered during preprocessing, including consistent resizing and basic quality control.

The output page shows successful visual outputs for five patients: Patient 1, Patient 10, Patient 29, Patient 5, and Patient 6.

4. Task 2 – Signal Processing: Pulse Estimation

The second part of the project focuses on pulse detection from forehead and sclera video data. The binary files were decoded into RGB video frames and converted into temporal signals by analyzing pixel intensity changes over time.

The general signal processing pipeline includes:

I. Reading binary video files
II. Reshaping the data into video frames
III. Extracting RGB/color-channel signals
IV. Normalizing the temporal signal
V. Applying band-pass filtering
VI. Performing frequency-domain analysis
VII. Estimating the dominant pulse frequency
VIII. Converting the frequency value into BPM

5. Outputs

The project was completed successfully for both the computer vision and signal processing parts. The final output summary includes visual ROI results for Task 1 and BPM estimations for Task 2.

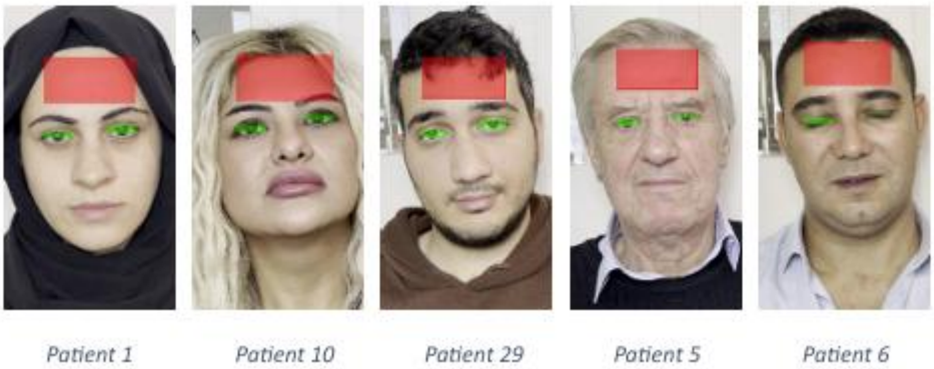

Figure 1: Task1 ROI Output

Patient ID	Forehead BPM	Sclera BPM
1	59.2	63.4
10	62.9	57.5
29	44.5	53.1
5	51.9	64.9
6	53.6	52.8

Table 1: Obtained BPM Values

These results show that the signal processing pipeline was able to produce numerical pulse estimations from both forehead and sclera video data.

6. Discussion

The project demonstrates a complete workflow from visual region extraction to physiological signal estimation. The computer vision component prepares meaningful facial regions, while the signal processing component extracts temporal intensity variations and estimates pulse rate from those signals.

The main strength of the project is that it combines two different technical areas: computer vision and biomedical signal processing. The ROI extraction step provides structured input data, and the signal processing step transforms this data into interpretable physiological measurements.

Possible improvements include:

- More robust landmark-based ROI tracking
- Motion artifact reduction
- Comparison of RGB channels and chrominance-based methods
- Use of ICA/PCA-based signal separation
- Validation against ground-truth heart rate values
- More detailed visualization of time-domain and frequency-domain signals

7. Conclusion

This project implemented a two-stage pipeline combining computer vision and biomedical signal processing for non-contact pulse estimation. First, facial regions of interest were extracted from patient video data and stored in binary format. Then, forehead and sclera video signals were processed to estimate pulse rate by analyzing subtle pixel intensity variations over time.

The approach follows the general principles of remote photoplethysmography, where small color changes in facial regions are used to infer cardiac activity. The implementation includes core steps commonly used in rPPG studies, such as ROI-based signal extraction, color-channel analysis, filtering, and frequency-domain pulse estimation. The obtained BPM values show that meaningful pulse estimates can be produced from both forehead and sclera regions. However, since no ground-truth heart rate data was available, the results should be considered experimental rather than clinically validated. Video-based pulse estimation is also sensitive to illumination, motion artifacts, ROI stability, camera quality, and preprocessing choices.

Overall, the project demonstrates a complete workflow from facial ROI extraction to pulse estimation and provides a practical foundation for further work in remote physiological monitoring. Future improvements could include ground-truth validation, more robust ROI tracking, and comparison with methods such as CHROM, POS, PCA/ICA, or learning-based rPPG approaches.